

Tri-Modal Sequence Model (TSM): Hybrid AI Architecture

Research Focus

This research investigates the computational limitations of Transformer self-attention, which scales as $O(n^2)$. The goal is to design sub-quadratic architectures by rethinking attention and introducing structured, persistent, and hierarchical memory systems.

Project Objective

To build a hybrid AI architecture combining Transformer (attention), Mamba (state space), and Hyena (convolution) to create an efficient and scalable sequence model.

Project Name

Tri-Modal Sequence Model (TSM)

Core Idea

The model integrates three memory paradigms: Mamba for long-term persistent memory, Transformer for precise relational reasoning, and Hyena for global pattern extraction.

Architecture

Input → Embedding → Parallel Branches (Mamba | Transformer | Hyena) → Fusion Layer → Feedforward → Output

Memory Roles

- 1 Mamba: Long-term memory through state evolution
- 2 Transformer: Short-term relational reasoning
- 3 Hyena: Global pattern recognition via convolution

Expected Outcomes

- 1 Working tri-hybrid prototype model
- 2 Performance comparison across paradigms
- 3 Understanding of memory efficiency vs accuracy trade-offs
- 4 Foundation for novel AI architectures

Innovation

This project explores a unified memory system rather than relying on a single mechanism, aligning with emerging trends in next-generation AI architectures.

Future Scope

- 1 Adaptive gating between components
- 2 Hierarchical memory layers
- 3 Scaling to long-context tasks
- 4 Potential research publication